TECHNICAL & BUSINESS OVERVIEW

# QuantFolio

A portfolio risk, performance and attribution platform built for Indian markets first.

| 62,645 | 4,365 | 92% | 7,773 | 7 |
|---|---|---|---|---|
| LINES OF SOURCE | TESTS PASSING | TEST COVERAGE | INSTRUMENTS | DEPENDENCIES |

| | |
|---|---|
| **Application** | QuantFolio v2.6.0 |
| **Stack** | Python 3.11+ · Streamlit · pandas / NumPy / SciPy · Plotly · SQLite |
| **Coverage** | Full NSE + BSE equity listing, US large caps, global indices |
| **Deployment** | Runs locally or in Docker; no server, no database service, no API key |

> **What this document is.** Part 1 explains what the system does and the finance behind it. Part 2 covers the engineering. Part 3 is an honest list of what it does *not* do. Part 4 is a prioritised roadmap of what would make it more valuable to you or to a company.

Generated 11 September 2026 · market data as of 2 September 2026
All figures in this document were measured from the codebase, not estimated.

## 1 · What the system does

QuantFolio turns a list of holdings — typed in, or imported unmodified from a broker CSV — into a full analytical picture of a portfolio: what it is worth, what return it earned, how much risk it carries, where that risk comes from, and what would happen to it under stress.

It is organised as **16 pages** across four areas:

| Area | Pages | What they answer |
|---|---|---|
| **Portfolio** | Build · Overview · Holdings · Performance · Trades & lots | What do I own, what is it worth, what did it actually return, and what are my realised gains? |
| **Risk** | Risk analytics · Benchmark · Correlation & factors · Stress testing | How much can I lose, how do I compare to an index, what drives my returns, and what happens in a crash? |
| **Strategy** | Optimiser · Monte Carlo · Rebalance | What mix would have been better, what range of outcomes is plausible, and what trades get me there? |
| **Research** | Security deep dive · Watchlist · News & calendar · Export | How healthy is this company, what am I tracking, what is happening today, and how do I share it? |

### The features that distinguish it

- **Search by company name** across the whole Indian market — type *Reliance* or *HDFC Bank*, no need to know that the symbol is `RELIANCE.NS`. Typos are tolerated. Runs offline.
- **Broker CSV import** that works on a Zerodha, Groww or Upstox export *unmodified* — column names are recognised automatically, and every rejected row is reported rather than silently dropped.
- **Scheduled email tearsheets** — daily, weekly or monthly, delivered by a command your operating system schedules.
- **Watchlist price alerts** that e-mail you once when a level is reached, not once per check.
- **News & economic calendar** — live headlines matched to *your* holdings, plus the week's high-impact releases.
- **Fundamental quality scores** computed from a company's own financial statements.

## 2 · The finance knowledge used

This is the substance of the product. Every calculation below is implemented from its original definition, with the convention stated in the code and in the interface.

### 2.1 Return and performance measurement

| Concept | What it measures and why it is here |
|---|---|
| **Time-weighted return (TWR)** | Return of the *strategy*, stripping out the effect of deposits and withdrawals. The correct basis for comparing against an index. |
| **Money-weighted return (XIRR)** | Return *you* actually earned, including the timing of your contributions. Solved numerically and verified by checking that the net present value of the cash flows is zero at the answer. |
| **Modified Dietz** | A cash-flow-weighted approximation for periods without daily valuations. |

| Concept | What it measures and why it is here |
|---|---|
| **FIFO tax lots** | Every buy is a lot; every sell consumes the oldest lots first. This produces realised profit and loss, holding periods and an accurate cost basis. |
| **Geometric annualisation** | Compounding, not averaging — 252 trading days a year. |

### 2.2 Risk measurement

| Concept | What it measures and why it is here |
|---|---|
| **Volatility & downside deviation** | Total variability, and variability of losses only. |
| **Value at Risk (Cornish–Fisher)** | Loss threshold at a confidence level, *adjusted for skew and fat tails*. Real returns are not normally distributed and a normal VaR understates the danger. |
| **Conditional VaR** | The average loss *given* that the VaR threshold was breached — what the bad days actually cost. |
| **Kupiec backtest** | A statistical test of whether the VaR model was right, by counting how often it was breached. Rare in retail tools; it is what separates a risk number from a risk *model*. |
| **Maximum drawdown & episodes** | Worst peak-to-trough fall, plus every episode with its depth, length and recovery time. |
| **Sharpe · Sortino · Calmar · Omega · Ulcer** | Five risk-adjusted return ratios, each penalising a different definition of risk. |
| **Beta, alpha, tracking error, information ratio** | Market sensitivity, skill after adjusting for it, and consistency of the difference. |

### 2.3 Attribution — where risk and return come from

| Concept | What it measures and why it is here |
|---|---|
| **Marginal & component risk contribution** | How much of total portfolio volatility each holding is responsible for — which is *not* the same as its weight. A 5% position can carry 20% of the risk. |
| **Herfindahl index & effective number of bets** | Concentration. Twenty holdings can behave like four if they move together. |
| **Principal component analysis** | How many independent sources of risk actually exist in the portfolio. |
| **Hierarchical clustering** | Groups holdings by how they co-move, revealing hidden concentration that sector labels miss. |
| **Fama–French factor regression** | Decomposes returns into market, size, value, profitability and momentum exposures — with **Newey–West standard errors**, because return data is autocorrelated and ordinary errors would overstate significance. |

### 2.4 Portfolio construction

| Concept | What it measures and why it is here |
|---|---|
| **Mean–variance optimisation** | Maximum-Sharpe and minimum-variance portfolios, and the efficient frontier, solved by sequential quadratic programming. |

| Concept | What it measures and why it is here |
|---|---|
| **Covariance shrinkage** | A sample covariance matrix estimated from limited history is unstable; shrinkage pulls it toward a structured target so the optimiser does not act on noise. |
| **Risk parity** | Weights that equalise *risk* contribution rather than capital. |
| **Black–Litterman** | Starts from the market's own implied returns (reverse optimisation) and blends in your views with a confidence level. Solves the classic problem that raw mean–variance produces absurd, unstable weights. |
| **Monte Carlo simulation** | Thousands of forward paths to show a *distribution* of outcomes instead of a single projection. |
| **Rebalancing trade lists** | The actual buys and sells to reach a target, with drift tolerance bands. |

### 2.5 Indian-market specifics

| Concept | What it measures and why it is here |
|---|---|
| **STCG / LTCG capital gains** | Indian equity tax under Sections 111A and 112A, including the **23 July 2024 regime change** (15%→20% short term, 10%→12.5% long term, exemption ₹1L→₹1.25L). |
| **Loss set-off & carry-forward** | Sections 70/71/74 — set-off is *mandatory in quantum* and only elective in ordering, and losses carry forward eight years. |
| **Section 112A grandfathering** | The 31 January 2018 cost-basis rule. |
| **Tax-loss harvesting** | Which lots to realise, and the tax actually saved — which is often zero, and the tool says so. |
| **Exchange conventions** | `.NS` / `.BO` suffixes, NSE series (EQ/BE/BZ), BSE groups, ISIN (ISO 6166) with check-digit validation. |

### 2.6 Derivatives and company analysis

| Concept | What it measures and why it is here |
|---|---|
| **Black–Scholes–Merton** | European option pricing with all five Greeks, and implied volatility solved by root-finding. NSE options *are* European, so the model is exactly right for this market. |
| **Covered calls, protective puts, collars** | Payoff, breakeven, income, and the upside surrendered — always plotted *against simply holding the shares*. |
| **Piotroski F-Score** | Nine-signal test of fundamental strength (Piotroski, 2000). |
| **Altman Z-Score** | Distress risk, using the Z″ variant and the emerging-market adjustment where appropriate (Altman 1968, 2005). |
| **Beneish M-Score** | Eight-variable screen for aggressive accounting (Beneish, 1999). |
| **Sloan accruals ratio** | How much of reported profit is not backed by cash (Sloan, 1996). |
| **Technical indicators** | Wilder RSI, MACD, Bollinger bands, 52-week positioning. |

**A principle that runs through all of it.** Where a number cannot be computed honestly, the system says so rather than substituting a plausible value. A missing 52-week range shows as a dash, not 50%. An interest-cover ratio for a company reporting no interest expense is *unavailable*, not infinite. A Piotroski score built from six of nine available signals is reported as "six of eight passed", never as a low nine — because counting a missing signal as a failure makes a healthy company look distressed.

## 3 · The programming knowledge used

### 3.1 Languages, libraries and tools

| Tool | Role | Why this one |
|---|---|---|
| **Python 3.11+** | Language | Modern typing, dataclasses, pattern of the scientific stack. |
| **pandas / NumPy** | Data & maths | Time-series alignment across exchanges and time zones; vectorised computation. |
| **SciPy** | Optimisation & statistics | SLSQP for the optimiser, Brent root-finding for XIRR and implied volatility, chi-square for the VaR backtest, hierarchical clustering. |
| **Streamlit** | Web interface | Full interactive application from pure Python — no JavaScript, no separate front end to maintain. |
| **Plotly** | Charts | 61 interactive chart builders on one shared theme. |
| **yfinance** | Market data | Free, broad coverage, and honest about being delayed. |
| **SQLite** (standard library) | Persistence | One file, no server. Saved portfolios, watchlists, alert rules and schedules. |
| **XlsxWriter** | Excel export | Formatted multi-sheet workbooks with live formulas. |
| **pytest · ruff · mypy** | Quality | Testing, linting/formatting, static type checking. |
| **Docker · Make · GitHub Actions** | Delivery | Reproducible builds, one-command tasks, CI on Linux and Windows. |

**Seven runtime dependencies, deliberately.** Features that would normally pull in new libraries were built on the standard library instead: the exchange-file downloader and news reader use `urllib` (no `requests`), the e-mail sender uses `smtplib`, scheduling uses `zoneinfo`. Fewer dependencies means fewer security advisories, faster installs and less that can break.

### 3.2 Software architecture

The codebase follows two rules, enforced consistently:

> **1. Computation never imports the user interface. The interface never computes.**
> 21 analytics modules contain zero Streamlit imports; the UI contains zero formulas. This is why ~20,000 lines of financial mathematics can be tested in seconds, offline, against hand-computed values — with no browser and no market-data subscription.

> **2. Every network call lives in one package.**
> Three modules in `src/data/` touch the wire — Yahoo prices, the exchange files, the news feeds — and nothing outside it does. Replacing the price provider with a broker feed (Zerodha Kite, Upstox, Interactive Brokers) means rewriting *one file* against the same signatures: no change to the analytics, the interface, or the tests.

| Layer | Lines | Responsibility |
|---|---|---|
| `src/analytics/` | 19,512 | 21 pure modules — all the financial mathematics |
| `src/ui/` | 19,533 | Design system, session state, 16 pages |
| `src/charts/` | 8,315 | 61 Plotly builders on one theme |

| Layer | Lines | Responsibility |
|---|---|---|
| `src/reporting/` | 5,718 | Excel, HTML tearsheet, SMTP delivery, scheduling |
| `src/data/` | 3,342 | Market data, exchange files, news, SQLite store |
| **Tests** | 36,068 | 36 modules — more test code than application code |

### 3.3 Engineering techniques demonstrated

- **Test-driven correctness through invariants.** Rather than only checking fixed values, the suite asserts mathematical identities that cannot rot: beta of a series against itself is exactly 1.0; component risk contributions sum to portfolio volatility; CVaR ≤ VaR at every confidence level; put–call parity holds to $1\times10^{-12}$; every option Greek matches its numerical derivative.
- **Cross-module agreement.** Two independent implementations must produce the same answer — a one-factor regression beta equals the covariance/variance beta to $1\times10^{-12}$.
- **Headless UI testing.** All 16 pages are rendered and asserted in CI with a stubbed data layer — no browser, no network.
- **Typed interfaces.** Frozen dataclasses throughout; static type checking in CI.
- **Defensive data handling.** Column-name aliasing, encoding detection, timezone normalisation, schema contracts on every download, and atomic file writes with backups.
- **Caching strategy.** Tiered time-to-live by data type — 5 minutes for quotes, 30 minutes for history, 24 hours for fundamentals.
- **Command-line tooling.** Scheduled reports and the universe rebuild run headlessly under cron or Windows Task Scheduler, sharing the exact same computation path as the interface.

### 3.4 Data engineering

The instrument universe is **generated**, not hand-maintained. A build step pulls four official exchange files and merges them:

| Source | Rows | Supplies |
|---|---|---|
| NSE equity listing | 2,567 | Symbol, company name, series, ISIN |
| BSE scrip master | 4,991 | Trading symbol, issuer name, ISIN, industry |
| NSE bhavcopy | 3,633 | Daily turnover — used to rank search results |
| BSE bhavcopy | 5,002 | Daily turnover |

Turnover is not decoration. Without it a full-market search ranks alphabetically: typing *hdfc* returned HDFC AMC ahead of HDFC Bank, and the typo *relaince* returned Reliance *Communications* (₹0.3 crore traded) ahead of Reliance Industries (₹1,767 crore). Ranking by traded value fixed both. The build also refuses to keep a symbol that two independent sources agree has stopped trading — it caught four, including `TATAMOTORS.NS` after the group's demerger.

## 4 · What it deliberately does not do

Stated plainly, because a tool's credibility rests on its limits being known rather than discovered.

| Limitation | Why, and what it would take to remove |
|---|---|
| **Prices are delayed, not live** | Yahoo Finance runs roughly 15 minutes behind for NSE/BSE. Every price carries a timestamp and a freshness label; the application never claims to be real-time. *Removing it:* a broker feed such as Zerodha Kite or Upstox, behind the existing data interface. |
| **No long-horizon return forecast** | Predicting a 5-, 10- or 11-year return requires pairs of (accounts at year T, return from T to T+N). The data provider supplies five annual periods, which yields four one-year pairs and **zero** from five years out. Rather than print a number fitted to no observations, the application lists every horizon with its training-pair count and the reason. *Removing it:* a point-in-time fundamentals database with 15+ years of history retaining delisted companies (Compustat, CMIE Prowess, Refinitiv). This is a paid-data problem, not a modelling one. |
| **No option chain for NSE/BSE** | Measured, not assumed: Yahoo returns 21 expiries for `AAPL` and zero for `RELIANCE.NS`. Options analytics therefore work from a strike and premium typed off your broker screen. *Removing it:* an NSE data subscription. |
| **The app cannot schedule itself** | Streamlit runs only while a browser is connected. Scheduled e-mails are delivered by a command run by cron or Windows Task Scheduler — and the interface says so rather than showing a toggle that does nothing. |
| **Headline tone is not a prediction** | A small documented lexicon reads the *words*, and always shows which ones matched. It does not know what the market had already priced in. |
| **Insights are rules, not advice** | Every insight comes from a documented threshold. The application is analytical and educational; it does not make recommendations. |

> **Why this list is an asset, not a weakness.** Most retail analytics tools fail silently — a missing value becomes a zero, a stale feed keeps rendering, a model extrapolates past its data. This system was built so that each of those is caught and named. One live example: the news feeds originally specified still return valid data with a *200 OK* response, but have not published since April 2024. The application measures the age of every feed and withholds any source older than seven days — so it shows nothing rather than presenting 870-day-old headlines as today's market news.

## 5 · Where to take it next

Ordered by value relative to effort. `QUICK WIN` is days of work, `MEDIUM` is a week or two, `STRATEGIC` is a larger commitment or a paid data source.

### 5.1 Highest value for a personal investor

| Feature | Effort | Why it matters |
|---|---|---|
| **Sector data for NSE listings** | `QUICK WIN` | BSE publishes an industry label and NSE does not, so ~2,300 NSE-only instruments currently have a blank sector. This silently weakens sector attribution and concentration analysis — the two views most people look at first. |

| Feature | Effort | Why it matters |
|---|---|---|
| **News impact event study** | QUICK WIN | Measure the *realised* abnormal return around a stock's past news and results dates. This is the defensible answer to "what does news do to this stock", and the data pipeline for it already exists. |
| **Capital-gains statement export** | QUICK WIN | The FIFO lot engine and the Indian tax rules are already built. Producing a ready-to-file schedule of realised gains turns a feature into an annual reason to open the app. |
| **Dividend and corporate-action tracking** | MEDIUM | Splits and bonuses currently rely on adjusted prices. Explicit handling gives accurate income yield and a correct cost basis. |
| **Goal-based planning** | MEDIUM | "Will this portfolio fund a ₹2 crore goal in 12 years?" The Monte Carlo engine already produces the distribution; this frames it as a probability of success. |
| **Mutual fund & SIP support** | STRATEGIC | Most Indian retail wealth is in mutual funds, not direct equity. AMFI publishes daily NAVs free. This is the single largest expansion of the addressable audience. |

### 5.2 Highest value for a company or an advisory practice

| Feature | Effort | Why it matters |
|---|---|---|
| **Multi-client / multi-account view** | MEDIUM | The store already holds multiple named portfolios. Roll-ups across clients, with per-client reports, turn a personal tool into an advisory product. |
| **Branded PDF client reporting** | QUICK WIN | The HTML tearsheet and the scheduled e-mail already exist. Adding a logo, a cover page and an adviser footer makes it a client deliverable. |
| **User accounts & access control** | MEDIUM | Required before any multi-user deployment. Streamlit supports authentication; the data layer would move from SQLite to PostgreSQL. |
| **Audit trail** | MEDIUM | Who changed what, and when. Necessary for any regulated context, and straightforward given the existing store. |
| **Broker API integration** | STRATEGIC | Zerodha Kite or Upstox would remove manual import entirely and provide real-time prices. The architecture already isolates this to one module. |
| **Benchmark & peer comparison** | MEDIUM | Compare a client portfolio against model portfolios or a peer group, not only an index. |
| **Hosted deployment** | STRATEGIC | The Docker image exists. A hosted instance with scheduled reports makes it a service rather than a program. |

### 5.3 Analytical depth, if differentiation is the goal

- **Point-in-time fundamentals** — the one purchase that would unlock genuine multi-year return modelling, cleanly and honestly.
- **Custom factor construction** — build and backtest factors on the full NSE/BSE universe; the regression machinery and the instrument data are already in place.
- **Transaction-cost and slippage modelling** — rebalancing currently ignores brokerage, STT and impact cost, which materially changes whether a trade is worth making.

- **Fixed income and gold** — most Indian portfolios are multi-asset; equity-only analysis tells a partial story.
- **Scenario builder** — user-defined shocks ("rupee falls 10%, oil rises 30%") on top of the existing historical stress replay.

> **If only three things are done next:** ① NSE sector data, because it silently degrades analysis people already rely on; ② the capital-gains statement, because it creates a recurring reason to use the product; ③ mutual fund support, because it is where most Indian household wealth actually sits.

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**QuantFolio v2.6.0** · 62,645 lines of source · 36,068 lines of tests · 4,365 tests passing · 92% coverage · 7,773 instruments · 5,175 ISIN mappings

Educational and analytical software. It does not provide investment advice, and it does not claim to supply real-time market data.